

MANASSAS, VA, USA

WMO: 724036

Lat: 38.721N

Lon: 77.515W

Elev: 59

StdP: 100.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.6	(c) -8.9	(d) -18.8	(e) 0.7	(f) -8.7	(g) -16.4	(h) 0.9	(i) -6.2	(j) 11.5	(k) -0.9	(l) 10.6	(m) -0.5	(n) 1.3	(o) 330	(p) 0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	33.9	23.8	32.7	23.6	31.4	23.0	25.6	31.5	24.8	30.3	24.1	29.3	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.9	28.3	23.0	17.9	27.4	22.5	17.3	26.9	78.5	31.6	75.4	30.1	72.5	29.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.3	7.3	-16.4	35.8	3.4	1.4	-18.9	36.8	-20.8	37.6	-22.7	38.4	-25.2	39.5
			-17.2	26.6	4.3	1.0	-20.3	27.3	-22.9	27.9	-25.3	28.4	-28.4	29.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	0.5	2.8	6.3	12.8	17.3	22.3	24.6	23.9	20.3	13.7	7.1	2.2
	DBStd	9.56	5.88	5.23	4.96	4.85	4.38	3.38	2.60	2.75	3.61	4.62	4.62	4.73
	HDD10.0	1043	301	207	138	24	1	0	0	0	0	14	112	246
	HDD18.3	2640	552	434	374	177	75	8	1	1	22	159	338	499
	CDD10.0	2086	7	7	23	108	226	368	452	429	308	128	24	5
	CDD18.3	641	0	0	1	11	42	126	194	172	80	14	1	0
	CDH23.3	6204	0	3	22	174	444	1214	1906	1564	729	139	7	2
	CDH26.7	2427	0	0	5	49	146	471	841	637	247	32	1	0
Wind	WSAvg	2.5	2.9	2.7	3.4	3.2	2.5	2.3	2.0	1.9	2.2	2.2	2.4	2.5
Precipitation	PrecAvg	1048	75	55	97	83	103	101	103	83	104	80	81	81
	PrecMax	1433	168	166	229	126	208	213	180	185	259	205	157	146
	PrecMin	746	25	15	0	40	35	35	34	31	0	2	22	40
	PrecStd	157	40	31	50	27	47	45	41	36	66	51	41	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.4	26.3	30.9	32.7	34.1	36.2	35.2	33.9	29.5	23.9	19.9
		MCWB	15.5	15.5	16.0	19.9	22.4	24.1	24.8	24.2	22.4	22.2	17.0	14.3
	2%	DB	17.4	17.6	22.2	27.9	30.2	32.5	34.0	33.6	31.7	27.2	21.1	16.2
		MCWB	13.7	11.3	13.4	17.9	21.4	23.4	24.2	23.9	22.0	19.6	15.6	11.7
	5%	DB	13.8	14.0	18.7	25.0	27.9	31.2	32.6	32.2	29.3	24.1	18.0	12.8
		MCWB	10.2	8.4	12.1	16.6	20.4	23.0	24.0	23.6	21.3	18.4	13.7	9.0
	10%	DB	9.0	11.3	16.0	22.4	25.9	29.5	31.3	30.5	27.5	22.3	16.0	10.1
		MCWB	5.9	7.2	9.9	15.2	18.8	22.2	23.4	23.0	19.8	17.5	11.4	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	16.8	17.2	21.3	24.4	25.5	26.8	26.3	25.4	23.7	18.6	15.7
		MCDB	18.6	21.3	24.3	26.9	30.5	32.0	33.3	32.6	30.0	27.6	21.5	18.5
	2%	WB	14.3	12.5	15.0	19.6	22.7	24.6	25.7	25.2	23.8	22.1	17.0	12.7
		MCDB	17.0	16.8	20.6	25.6	28.1	30.5	31.9	30.8	28.0	24.8	20.0	15.5
	5%	WB	10.5	9.8	12.7	17.9	21.4	23.8	25.0	24.5	22.7	20.3	14.5	9.6
		MCDB	13.6	12.5	17.5	23.3	26.5	29.4	30.6	29.8	26.9	23.3	17.4	12.7
	10%	WB	6.2	7.5	10.3	16.1	20.1	23.0	24.1	23.7	21.6	17.9	11.9	6.7
		MCDB	8.8	10.7	15.4	21.5	24.5	28.2	29.3	28.6	25.8	21.3	14.9	9.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	12.3	13.6	12.6	11.9	11.6	11.5	12.3	13.1	12.4	10.8
		MCDBR	15.4	16.4	17.9	18.1	15.5	13.8	13.5	13.7	15.2	15.3	16.1	15.1
		MCWBR	11.8	11.5	10.8	9.5	8.1	6.3	5.5	5.5	6.9	9.0	11.1	11.0
	5% WB	MCDBR	13.6	13.6	15.4	14.8	13.2	12.2	11.9	11.2	11.6	11.5	12.7	13.8
		MCWBR	11.2	10.5	10.5	8.6	7.2	6.0	5.3	5.1	6.1	7.4	10.0	10.8
Clear-Sky Solar Irradiance	taub		0.304	0.320	0.355	0.399	0.438	0.498	0.515	0.513	0.425	0.359	0.322	0.308
	taud		2.489	2.422	2.388	2.287	2.205	2.077	2.052	2.037	2.261	2.448	2.511	2.514
	Ebn at Noon		877	905	903	878	847	794	776	765	818	849	848	846
	Edh at Noon		79	96	110	129	143	163	166	164	124	93	77	72
All-Sky Solar Radiation	RadAvg		2.10	2.93	3.89	4.88	5.50	6.06	5.92	5.30	4.34	3.27	2.38	1.84
	RadStd		0.19	0.33	0.26	0.36	0.58	0.46	0.32	0.26	0.52	0.37	0.20	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+76

Nomenclature: See separate page